L'Hôpital's Rule

Numberbender | WORKSHEET

Name: ________________________________ Date: ____________ Score: _______

Apply L'Hôpital's Rule where applicable. Show all work.

CALCULUS I
Worksheet #47

L'Hopital's Rule:	Suppose that f(a)=g(a)=0, that f & g are differentiable on an open interval containing a, and that $g'(x) \neq 0$ on ; then $\lim_{x\to a}\frac{f(x)}{g(x)} = \lim_{x\to a}\frac{f'(x)}{g'(x)}$
Examples:	$\lim_{x\to 2}\frac{x^2-4}{x-2} = \frac{0}{0} \Rightarrow$ Use L'Hopital's Rule - Take derivative of numerator and denominator $\Rightarrow$ $\lim_{x\to 2}\frac{2x}{1} = \boxed{4}$
	$\lim_{x\to 1}\frac{x-1}{\sqrt{x}-1} = \frac{0}{0} \Rightarrow$ Use L'Hopitals: $\lim_{x\to 1}\frac{1}{\frac{1}{2\sqrt{x}}} = \lim_{x\to 1}\frac{2\sqrt{x}}{1} = \boxed{2}$
	$\lim_{x\to 2}\frac{x^2-4}{x+2} = \frac{0}{4} = \boxed{0} \Rightarrow$ You cannot use L'Hopital's Rule on this problem! Why??

Homework:

1	$\lim_{x\to 2}\frac{x^3-8}{x^2-4}$	2	$\lim_{x\to 0}\frac{\sin 5x}{x}$
3	$\lim_{x\to 2}\frac{\sqrt{2+x}-2}{x-2}$	4	$\lim_{x\to 1}\frac{\sqrt[3]{x}-1}{x-1}$
5	$\lim_{x\to 2}\frac{x^2-4x+4}{x^3-12x+16}$	6	$\lim_{x\to \frac{\pi}{2}}\frac{1-\sin x}{1+\cos 2x}$
7	$\lim_{x\to 1}\frac{x^3-1}{4x^3-x-3}$	8	$\lim_{x\to 3}\frac{x-4}{x-2}$
For 9-12: $\boxed{\int_{-1}^{0} f(x)dx = -2}$ $\boxed{\int_{0}^{1} f(x)dx = 1}$ $\boxed{\int_{-1}^{0} g(x)dx = 3}$ $\boxed{\int_{-1}^{1} g(x)dx = 7}$ $\boxed{\int_{-1}^{1} h(x)dx = 5}$ $\boxed{\int_{0}^{1} h(x)dx = 11}$			
9	$\int_{-1}^{1} f(x)dx$	10	$\int_{-1}^{0} h(x)dx$
11	$\int_{-1}^{-1} g(x)dx$	12	$3\int_{-1}^{0} g(x)dx - 2\int_{-1}^{0} h(x)dx$
13	If f(x)= $x^2 \tan^{-1} x$, then f '(1) = ??	14	Find the area between the curve $y^2 = 4x$ and the line y=x.
15	Find the area of the region enclosed by $y = e^x$, $y = \frac{1}{x}$, x = 1 and x = 2.	16	$\int_{0}^{1} x\left(5^{-x^2}\right)dx$
17	$\int_{1}^{2} 2^x dx$	18	$\sum_{j=-3}^{3}(2j+1) =$

Answers:

Answer key — for instructor use only.

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1) 3	2) 5	3) $\frac{1}{4}$	4) $\frac{1}{3}$
5) $\frac{1}{6}$	6) $\frac{1}{4}$	7) $\frac{3}{11}$	8) -1
9) -1	10) -6	11) 0	12) 21
13) $\frac{\pi+1}{2}$	14) $\frac{8}{3}$	15) $e^2-e-\ln 2$	16) $\frac{4}{10\ln 5}$
17) $\frac{2}{\ln 2}$	18) 7		